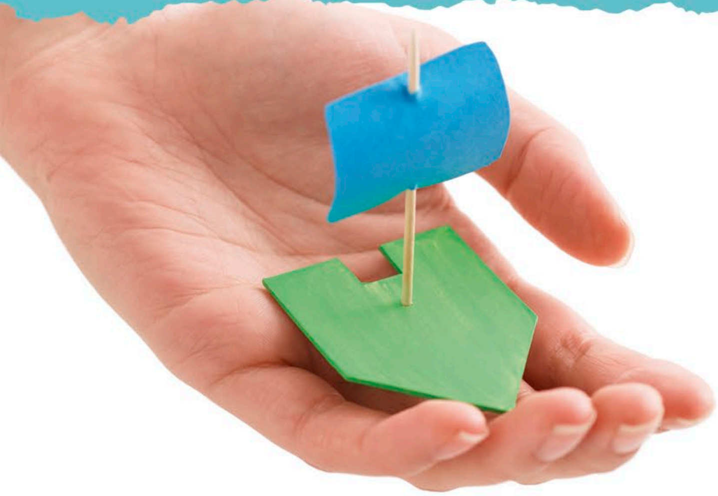
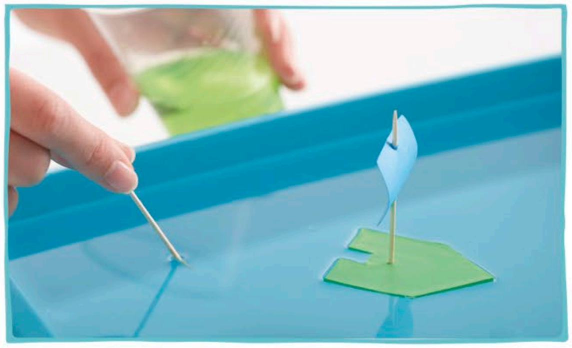


4 When the paint dries completely, it's time to secure the sail and toothpick to the boat. Your boat is ready to set sail!

5 Float your boat on the water-filled tray—keep it close to one corner and point it toward the middle. Dip a toothpick in dishwashing liquid and touch the water's surface in the square notch.



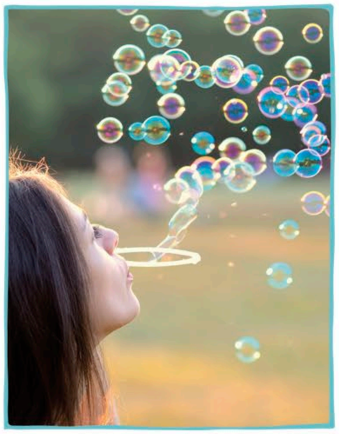
6 Watch your boat go! Just keep dipping the toothpick in the dishwashing liquid and touching it to the notch. But if the water gets too soapy, you'll need to change it or the experiment won't work.

HOW IT WORKS

This soapy experiment makes use of something called surface tension—because water molecules cling together, they pull each other in all directions. This motion pulls tight the surface of the water, like the skin of a balloon. But when you drop dishwashing liquid into the water, the bonds behind the boat weaken, reducing surface tension. As a result, the rest of the water surface pulls away, dragging the boat along with it.

Surface tension is due to invisible forces pulling water molecules together.

REAL WORLD SCIENCE
BLOWING BUBBLES



Have you ever wondered why you can't blow bubbles with just water? It's because the surface tension is so strong you can't make the water stretch into a different shape. Mixing in soap reduces the surface tension enough for you to blow air inside the water without the bubble collapsing immediately.